

Samridh Prabhakar

prabhak9@purdue.edu | (765)-694-9281 | linkedin.com/in/samridh-prabhakar

EDUCATION

Purdue University

Bachelor of Science in Computer Engineering, Certificate in Semiconductors and Microelectronics

Graduation: Dec 2027

GPA: 3.36

EXPERIENCES

Undergraduate Process Engineer

Purdue Vertically Integrated Projects, Semiconductors @ Birck

Jan 2026 – Present

West Lafayette, IN

- Fabricated metal-insulator-metal capacitors in a cleanroom environment following microfabrication protocols.
- Conducted photolithography process steps and monitored parameters to ensure process repeatability.
- Applied **Statistical Process Control (SPC)** techniques to analyze fabrication data and maintain process stability.

Research Assistant

MMH Labs, Purdue University

Oct 2025 – Present

West Lafayette, IN

- Integrated flash memory with a **Nordic nRF52-based board** for embedded data storage.
- Focused on the packaging and integration of these components into single platform
- Developed and testing SPI communication protocols and firmware for reliable data storage and device operation

Student Researcher

Purdue Vertically Integrated Projects, Virtual Twins @ Birck

Aug 2025 – Dec 2025

West Lafayette, IN

- Designed semiconductor equipment using virtual twin models for visualization on the **Apple Vision Pro**.
- Gained hands-on experience with VR-based cleanroom operations for semiconductor fabrication training.
- Performed motion analysis and validation of semiconductor equipment through **Dassault Systèmes CATIA**.

Research Intern

University of Delhi

Apr 2023 – Jan 2024

Delhi, India

- Conducted specification-based testing of **silicon microstrip sensors** for the Compact Muon Solenoid Phase-II Upgrade at CERN, ensuring compliance with performance and quality standards.
- Performed detailed electrical characterization by measuring parameters including leakage current, capacitance, and resistance to identify defects and validate sensor reliability for high-energy physics applications.

PROJECTS

Embedded Multiplayer Tetris System | C++, ESP32, SPI, ESP-NOW

- Designed and developed a handheld **ESP32-based embedded gaming device** to run a real-time Tetris game with responsive rendering
- Implemented **wireless multiplayer functionality** using ESP-NOW protocol, enabling two devices to sync gameplay and track competitive scores in real time
- Built a graphical interface on an SPI-driven display, rendering game state, score, and next-piece preview with optimized memory usage

Location-Based Navigation Tool | Python, Raspberry Pi Zero W, Google Maps APIs

- Built a portable navigation device that points to the nearest point of interest using real-time GPS, compass heading, and Google Places API with **Python**
- Displayed destination, name, distance and direction on an OLED screen for user friendliness.
- Implemented Wi-Fi-based geolocation fallback and request caching for smooth operation.

SKILLS

Programming: C, Python, MATLAB, Verilog

Hardware & Systems: Raspberry Pi, Embedded Systems, Nordic nRF52, PCB Design, CATIA, Oscilloscopes

Development Tools: Git, VS Code, Google Cloud Platform, KiCad

Certifications: Linux and Bash (Purdue IEEE), Semiconductor Manufacturing 101 (Purdue, UTA & Intel), MATLAB Onramp (Mathworks)